Experiment 1:

TITLE: Dataset Creation and Updating using File Handling Program.

Objective: How to store and retrieve dataset in table format using file handling programming.

Aim:

To create and update a dataset using a file handling program and import it into MySQL database.

Theory:

File handling allows us to read and write data into files. We can use Python/Java/C/C++ to create and update text or CSV files, which can then be imported into a MySQL table using the `LOAD DATA INFILE` command.

SQL Code:

```
LOAD DATA INFILE 'data.csv'
INTO TABLE student
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

Output:

The expected output confirms successful execution of SQL commands and matches the theoretical concept.

Result:

Successfully executed as per the aim of the experiment.

Exercise

Design a program (using C, C++, Java or Python) that performs Create, Read, Update, Delete record using file handling. The file store information about students in the following format:

Roll Number (unique identifier), Name, Age, and Department

All data should be stored in a text file (students.txt) where each record is stored on a new line. The initial dataset can be downloaded from Kaggle or other sources for any other records like Employee Dataset, HR Analytics Employee Dataset etc.

Tasks to be carried out in LAB

1. **Create a Record:**

   Add a new student record to the file. Ensure that the roll number is unique (no duplicates).
2. **Read Records:**

   Display all student records stored in the file in a tabular format. Allow searching for a record by roll number.
3. **Update a Record:**

Update the details of a student by their roll number. Ensure the updated data is saved in the file.

4. **Delete a Record:**

Delete a student record using their roll number. Ensure the file reflects the changes after deletion.

5. **Exit Program:**

Provide an option to exit the program.

Menu:

1. Add a Student Record
2. View All Records
3. Search a Record by Roll Number
4. Update a Record
5. Delete a Record
6. Exit

Enter your choice: